

AN ULTRA LOW POWER 10 TRANSISTOR 1 BIT MAGNITUDE COMPARATOR USING GATE DIFFUSION INPUT TECHNIQUE

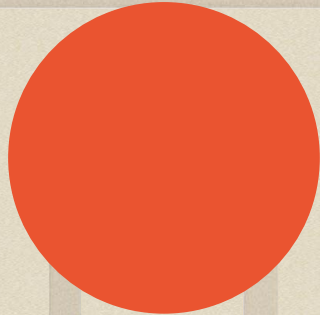
Shahrear Hossain Shawon

ID: ET223104

Supervised by:
Riazul Islam
Lecturer

Department of
Electrical and Electronic Engineering
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG

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INTRODUCTION

- Magnitude comparators are fundamental building blocks in digital VLSI systems, used to compare two binary inputs and determine A greater than B, A less than B, or A equal to B.
- They play a crucial role in arithmetic logic units (ALUs), processors, memory address decoding, digital controllers, and embedded systems.
- The growing demand for ultra-low-power and high-speed integrated circuits has increased the need for comparator designs with lower transistor count, reduced power consumption, and faster propagation delay
- This thesis proposes and analyzes an Ultra-Low-Power 10-Transistor 1-Bit Magnitude Comparator using the Gate Diffusion Input (GDI) Technique, and evaluates its performance in terms of power, propagation delay, and power-delay product (PDP) across 180 nm, 90 nm, and 45 nm PTM technology nodes. Select 52 more words to run Humanizer.

PROBLEM STATEMENT

- High transistor count in conventional CMOS comparators.
- Increased power consumption and silicon area.
- Higher propagation delay due to complex logic paths.
- Leakage current and parasitic capacitance at scaled technology nodes.
- Need for an ultra-low-power, high-speed comparator design.
- GDI technique offers reduced transistor count and improved efficiency.
- Comparative evaluation is required using Power, Delay, PDP, and Transistor Count.

OBJECTIVES

01

To design and simulate 1 Bit Comparator using CMOS GDI technique.

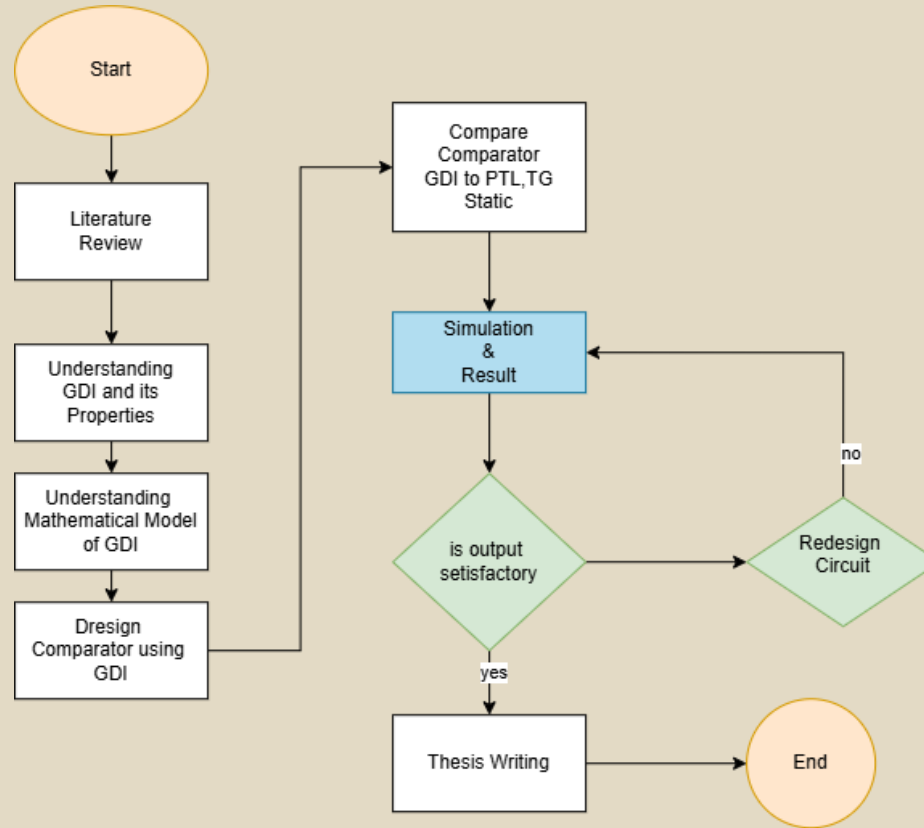
02

Reduce Transistor Number of the Comparator

03

To evaluate and compare key performance parameters such as: Rise time, Fall time, Propagation delay, Power consumption. PDP

METHODOLOGY



LITERATURE REVIEW

Researcher	Year	Contribution	Drawback
Arvind Kumar, Manoj Kumar	2017	Proposed a 1-bit CMOS comparator using transistor stacking across its XNOR and AND gates to reduce active-mode leakage power	Marginally higher propagation delay (49.38 ns) and increased transistor count due to splitting transistors into series pairs.
Dheeraj Tailor, Tavesh Sharma, Manoj Mumar	2019	Designed a 1-bit CMOS comparator utilizing MTCMOS (for NOR gates) and Forced Stacking (for inverters) to minimize subthreshold leakage and active power.	MTCMOS is a state-destructive technique (outputs from active mode are lost in standby mode); requires additional sleep control signal routing.

LITERATURE REVIEW

Researcher	Year	Contribution	Drawback
Samiha Lubaba, K. M. Faisal , Moumita Sadia Islam, Mehedi Hasan	2020	Developed a 2-bit hybrid magnitude comparator using PTL/TGL-based XNOR gates for low-latency internal signal generation combined with C-CMOS output logic for high driving power and full swing.	Requires a total of 46 transistors ; average power dissipation (7.792 μ W) remains slightly higher than pure GDI-based architectures
Yashmit Sharma, Premananda B.S, Divyanshu Ra, Kshitij Pandey	2025	Created a 1-bit hybrid comparator combining a 6T-MUX PTL Full Adder with TGL output gates , achieving a 22-transistor footprint and 58.7% power reduction	Pass-transistor pathways can exhibit threshold voltage drop unless carefully sized/driven; requires inverted input signals ($\bar{A}, \bar{B}$).

INTRODUCTION TO Comparator

- **Definition:** A digital magnitude comparator is a combinational logic circuit used to compare two binary numbers (A and B) and determine their relative numerical values.
- **Output Signals:** It generates three distinct output signals to indicate the exact comparison result: $A > B$ (Greater Than), $A < B$ (Less Than), and $A = B$ (Equal To).
- **Implementation Techniques:** Designed using various CMOS logic styles such as Complementary CMOS (C-CMOS), Transmission Gate Logic (TGL), Pass Transistor Logic (PTL), and Gate Diffusion Input (GDI).
- **Applications:** Used in ALU, processors, memory circuits, address decoding, and digital control systems.

CMOS Logic Styles

➤ Static Logic:

- Uses pMOS and nMOS in complementary fashion
- Stable output maintained without clock
- Low power dissipation when idle

➤ PTL Logic:

- Passes signals directly through single MOS switches to save area.
- Uses fewer transistors, lowering overall dynamic power usage.
- Loses voltage level (V_{th} drop) when passing a logic 1.

➤ TG Logic:

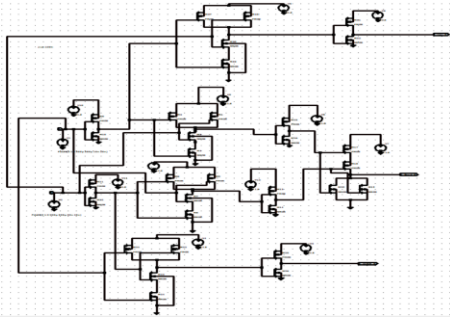
- Connects NMOS and PMOS in parallel to pass full logic levels
- Prevents voltage drop, giving clean strong 0 and 1 outputs.
- Needs inverted control signals, which increases

➤ GDI Logic:

- Uses a 3-input cell to implement complex functions in few transistors.
- Cuts down power and area for circuits like XOR and MUX.
- Needs special twin-well CMOS fabrication to avoid body effect issues.

CIRCUIT DESIGN COMPARE IN VARIOUS TECHNIQUE

Static CMOS (26T)



PROS:

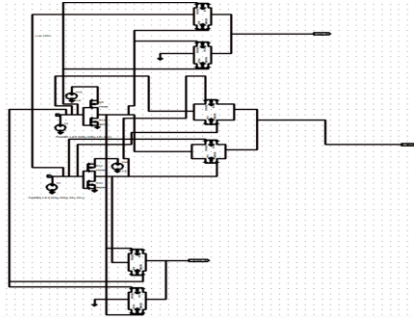
Full voltage swing (V_{DD} to GND), high noise margin.

CONS:

Excessive transistor count (26T), high parasitic capacitance.

Fig 01: Static CMOS

Transmission Gate (16T)



PROS:

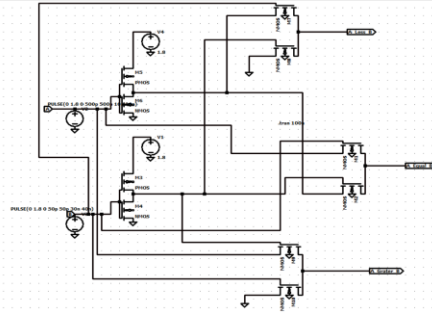
Full signal restoration, lower delay than CMOS.

CONS:

Requires complementary control signals (A , A'), high routing.

Fig 02: TG CMOS

Pass Transistor (12T)



PROS:

Reduced transistor count (12T-14T).

CONS:

Threshold voltage drop (V_{th} loss), signal degradation.

Fig 03: PTL CMOS

GATE DIFFUSION INPUT (GDI) TECHNIQUE

- **Basic Cell Structure:** Uses a 3-terminal cell—a common Gate (G), input to PMOS source/drain (P), and input to NMOS source/drain (N)—allowing inputs directly at diffusion terminals.
- **Key Advantages:** Realizes complex functions (XOR, MUX, AND, OR) with as few as two transistors, drastically reducing chip area, switching delay, and dynamic power consumption.
- **Limitations:** Requires custom silicon fabrication processes (like twin-well CMOS) or dedicated substrate connections to prevent threshold voltage degradation and body effect losses.

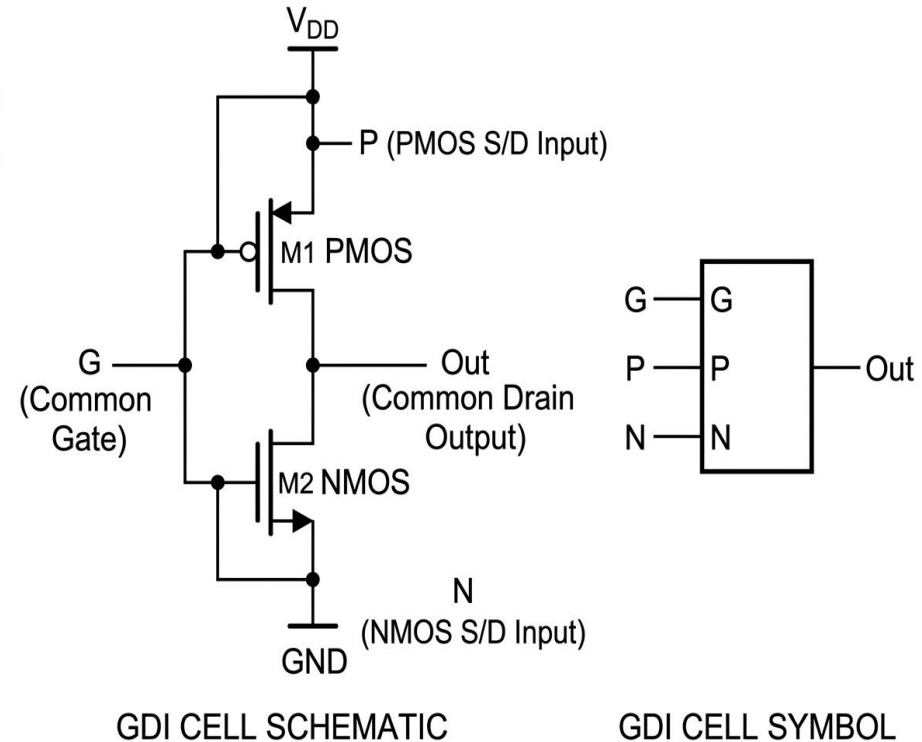


Fig 04: GDI CELL SCHEMATIC

PROPOSED 10 TRANSISTOR GDI MAGNITUDE COMPARATOR

Proposed 10 Transistor Schematic

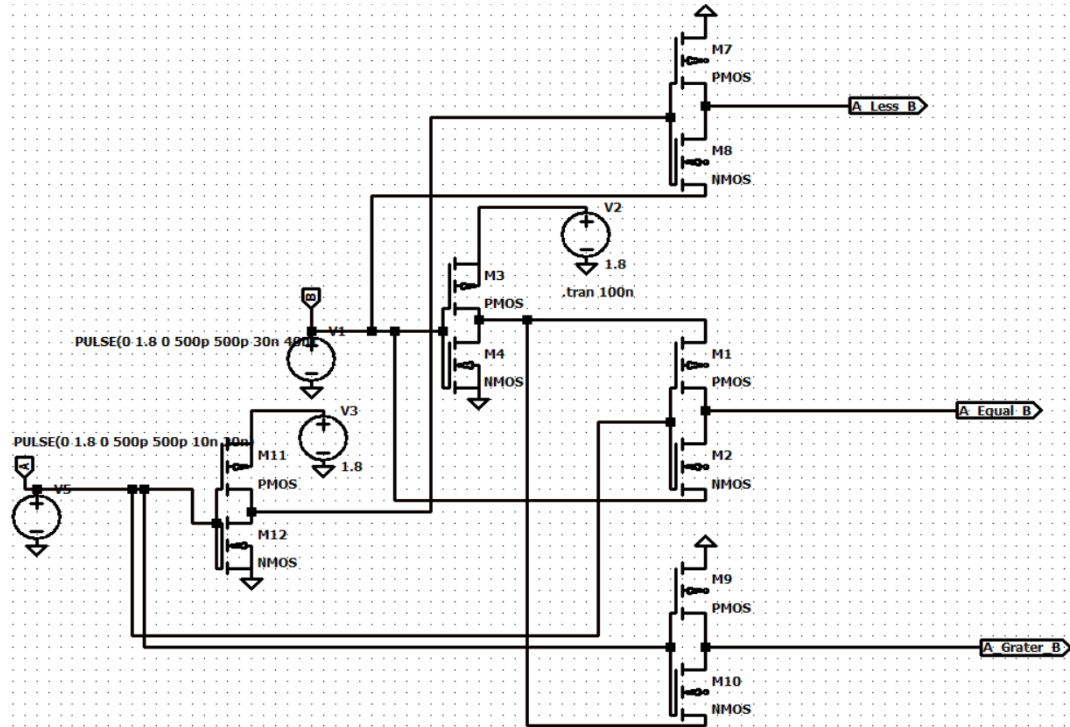


Fig 05: Proposed 10 Transistor Schematic

SIMULATION RESULTS

➤ Static CMOS Paper (2017) :

- Propagation Delay : 49.38 ns
- Average Power Dicipiation : 174.352 pW
- PDP: 0.0086f

➤ MTCMOS paper(2019):

- Propagation Delay : 330.61 ps
- Average Power Dicipiation : 48.12 pW
- PDP: 0.0000159f

➤ Static,Pass,TG paper (2020):

- Propagation Delay : 0.192 ns
- Average Power Dicipiation : 7.792 uW
- PDP: 1.496f

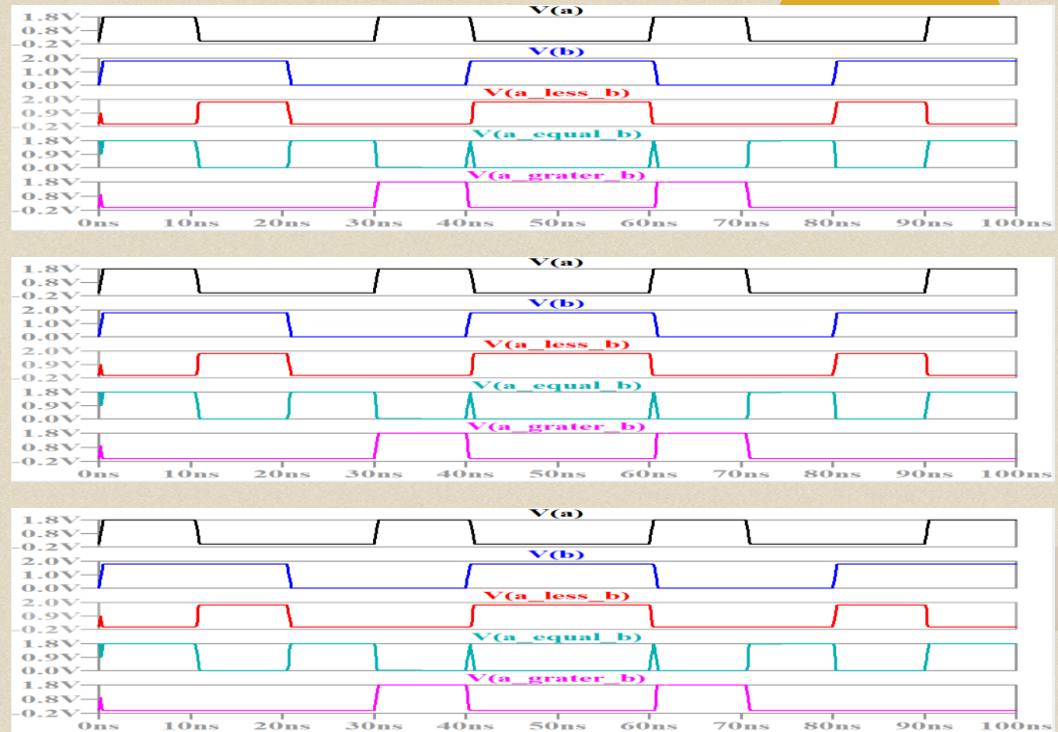


Fig 06 : Output Waveshape

SIMULATION RESULTS

➤ 6T MUX Adder Comparator(2025):

- Propagation Delay : 5.5 ps
- Average Power Dicipation : 7.3uW
- PDP: 0.04f

➤ MTCMOS paper(2019):

- Propagation Delay : 336 ps
- Average Power Dicipation : 34.18 pW
- PDP: 0.0000114f

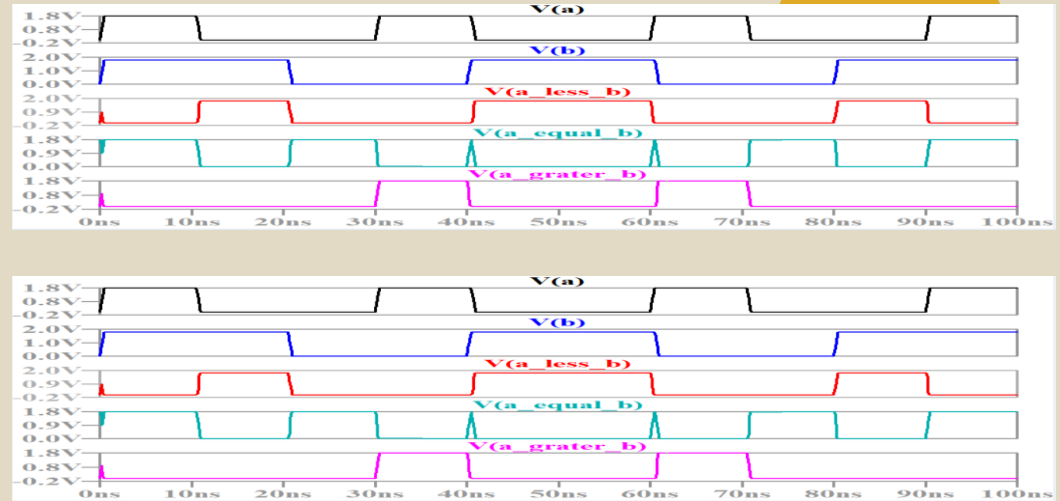


Fig 07 : Output Waveshape

PERFORMANCE COMPARISON

Propagation Delay Bar Chart

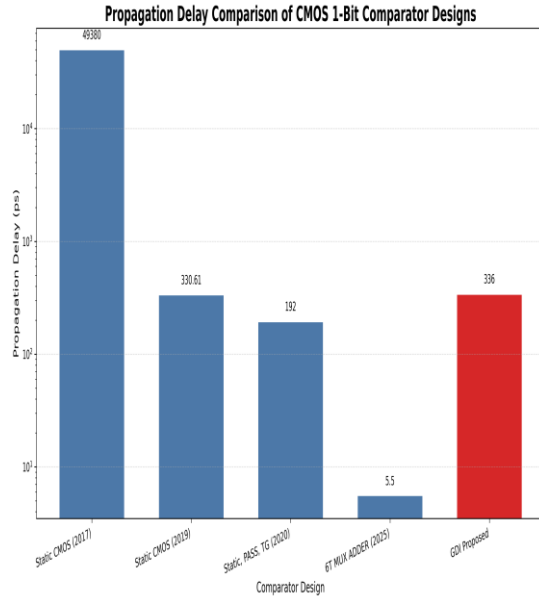


Fig 08 : Propagation Delay

Average Power Dissipation Bar Chart

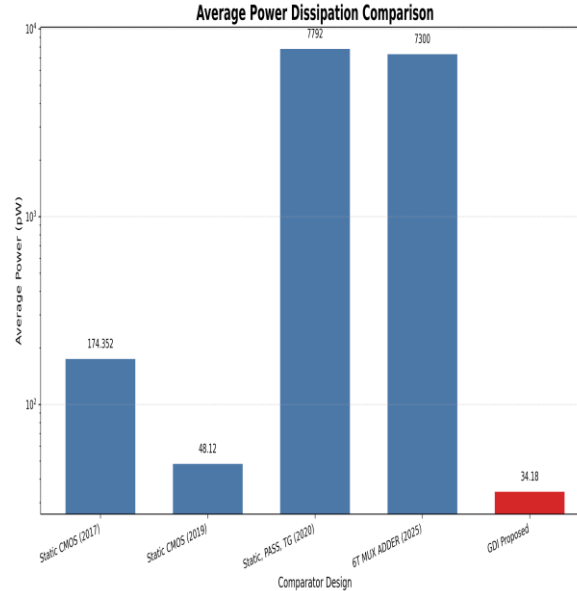


Fig 09 : Average Power Dissipation

PDP (Energy per Operation) Bar Chart

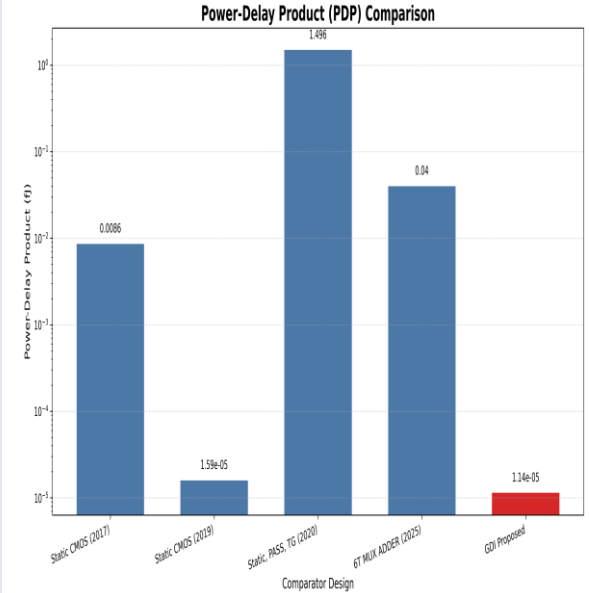
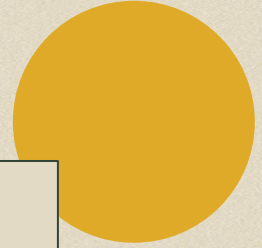


Fig 10: Product Delay Propagation

KEY FINDINGS



- GDI Technique demonstrate:
 - Its **low power consumption**.
 - Provides **short propagation delay**.
 - Requires **fewer transistors**.
- CMOS still viable for cost-sensitive applications

CONCLUSION

- CMOS faces scalability and performance limitations
- GDI technique addresses these with superior design
- For future VLSI design, GDI technique offers better performance and efficiency
- LTspice simulations confirm performance gains in GDI based designs

Future Work

- ❑ Study GDI Technique under varied temperatures and voltages
- ❑ Analyze integration into larger digital blocks (e.g., ALUs)
- ❑ Explore hybrid logic families for ultra-low power designs
- ❑ Investigate real-time implementations using FPGA or ASIC platforms
- ❑ Evaluate GDI performance in logic-intensive circuits like full adders or multiplexers
- ❑ Explore noise immunity and signal integrity analysis in GDI Technique
- ❑ Extend simulations to beyond to analyze scalability

REFERENCES

- [1] A. Kumar and M. Kumar, “Improved Design of CMOS 1-Bit Comparator with Stacking Technique,” *International Journal of Current Engineering and Technology*, Vol. 7, no. 3, pp. 1044–1047, 2017.
- [2] D. Tailor, T. Sharma, and M. Kumar, “Design of CMOS Based 1-bit Comparator with MTCMOS and Forced Stack Technique in 180 μ m Technology,” in *Proc. 2019 IEEE International Conference on Signal Processing and Communication (ICSC)*, NOIDA, India, pp. 330–334, 2019.
- [3] S. Lubaba, M. S. Islam, K. M. Faisal, and M. Hasan, “Design of a Two-Bit Magnitude Comparator Based on Pass Transistor, Transmission Gate and Conventional Static CMOS Logic,” in *Proc. 2020 IEEE Region 10 Symposium (TENSYP)*, Dhaka, Bangladesh, pp. 1114–1117, 2020.
- [4] Y. Sharma, B. S. Premananda, D. Raj, and K. Pandey, “Design and Analysis of a 1-bit Comparator for Low Power VLSI Systems,” in *Proc. 2025 Third International Conference on Networks, Multimedia and Information Technology (NMITCON)*, Bengaluru, India, pp. 1–6, 2025.

THANK
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